DeepResearch API

2.0.0 OAS 3.1

/openapi.json

An agentic research assistant API for discovering and exploring AI/ML academic papers. Features a 3-agent pipeline (classifier → query optimiser → BGE vector search) powered by Google Gemini, with `BAAI/bge-base-en-v1.5` cosine similarity search over a local arXiv AI/ML corpus (~521k papers). Summarisation via Gemini. Built with FastAPI, SQLAlchemy, ChromaDB, and SQLite. Authenticate via the `X-API-Key` header on all endpoints except `/health` .

Authorize

Search

Agentic search pipeline: classify the query, optimise it for retrieval, then perform BGE vector similarity search over a local arXiv AI/ML corpus.

GET `/search` Search for academic papers

Run the full agentic search pipeline: classify the arXiv AI/ML category, optimise the query, then perform BGE vector similarity search over the local arXiv AI/ML corpus.

Parameters

Try it out

Name	Description
query * required `string` *(query)*	Search query for academic papers query *minLength: 1*

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "field": "Machine Learning",
  "optimised_query": "key-value cache memory optimisati
on transformer language models",
  "original_query": "KV cache optimisation for transfor
mers",
  "result_count": 1,
  "results": [
    {
      "abstract": "The dominant sequence transduction m
odels...",
      "arxiv_id": "1706.03762",
      "authors": "Ashish Vaswani, Noam Shazeer",
      "categories": "cs.CL cs.LG",
      "similarity_score": 0.92,
      "title": "Attention Is All You Need",
      "url": "https://arxiv.org/abs/1706.03762",
      "year": 2017
    }
  ]
}
```


401	Missing or invalid API key	*No links*
422	Validation error — query parameter is required	*No links*
500	Vector search or Gemini agent failure	*No links*

Papers

Look up arXiv papers from the local corpus and find related papers via BGE vector similarity.

GET `/papers/{arxiv_id}` Get a paper from the corpus

Look up an arXiv paper by its ID from the local AI/ML corpus.

Parameters

Try it out

Name	Description
arxiv_id * required string (path)	arxiv_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. **Example Value** Schema	*No links*
401	Missing or invalid API key	*No links*
404	Paper not found in the corpus	*No links*
422	Validation Error	*No links*


```
{
  "abstract": "The dominant sequence transduction models are based on complex recurrent or convolutional neural networks...",
  "arxiv_id": "1706.03762",
  "authors": "Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin",
  "categories": "cs.CL cs.LG",
  "doi": "10.48550/arXiv.1706.03762",
  "similarity_score": 0.92,
  "title": "Attention Is All You Need",
  "url": "https://arxiv.org/abs/1706.03762",
  "year": 2017
}
```

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET /papers/{arxiv_id}/related Find related papers

Find papers related to the given arXiv paper using BGE vector similarity. Returns up to 5 related papers from the corpus.

Parameters

Try it out

Name	Description
arxiv_id * required string (path)	arxiv_id

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
[
  {
    "abstract": "The dominant sequence transduction mod
els are based on complex recurrent or convolutional neu
ral networks...",
    "arxiv_id": "1706.03762",
    "authors": "Ashish Vaswani, Noam Shazeer, Niki Parm
ar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukas
z Kaiser, Illia Polosukhin",
    "categories": "cs.CL cs.LG",
    "doi": "10.48550/arXiv.1706.03762",
    "similarity_score": 0.92,
    "title": "Attention Is All You Need",
    "url": "https://arxiv.org/abs/1706.03762",
    "year": 2017
  }
]
```


401 Missing or invalid API key *No links*

404 Paper not found in the corpus or vector store *No links*

422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


500 Vector search failure *No links*

Summarisation

AI-powered summarisation of academic text using the Gemini summariser agent. Rate limited to 5 requests/minute.

POST `/summarise` Summarise academic text

Summarise any academic text using the Gemini summariser agent. Pass an abstract for a quick preview, or the full paper text for a comprehensive summary. Provide arxiv_id to record this interaction in the community index.

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema

```
{
  "arxiv_id": "1706.03762",
  "text": "We propose a new simple network architecture, the Transformer, base
d solely on attention mechanisms, dispensing with recurrence and convolutions
entirely..."
}
```

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. **Example Value** Schema	*No links*

```
{
  "summary": "This paper introduces the Transformer, a
```

```
novel architecture that relies entirely on self-attenti
on mechanisms. It achieves state-of-the-art results on
machine translation benchmarks while being more paralle
lisable and requiring significantly less training tim
e."
}
```

401	Missing or invalid API key	*No links*
422	Validation error — text field is required and must not be empty	*No links*
429	Rate limit exceeded — max 5 requests per minute	*No links*
500	Gemini summarisation agent failure	*No links*

Community

Community activity index — discover papers trending by interaction count across the API.

GET `/community` List most popular papers

Returns papers ranked by community interaction count. A paper enters this index the first time it is directly accessed or summarised. Count increments on: direct paper lookup, summarisation, and related-papers queries.

Parameters

Try it out

Name	Description

limit
`integer`
`(query)`
	Number of papers to return (max 100)
	Default value : 20
	20
	maximum: 100
	minimum: 1

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "papers": [
    {
      "arxiv_id": "1706.03762",
      "authors": "Ashish Vaswani, Noam Shazeer",
      "categories": "cs.CL cs.LG",
      "interaction_count": 14,
      "last_interacted_at": "2025-03-10T11:42:00Z",
      "title": "Attention Is All You Need",
      "url": "https://arxiv.org/abs/1706.03762",
      "year": 2017
    }
  ],
  "total": 1
}
```


401	Missing or invalid API key	*No links*
422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

GET **/community/{arxiv_id}** Get community stats for a paper

Returns the interaction count and last activity timestamp for a specific paper. A paper only appears here once it has been accessed via GET /papers/{arxiv_id}, POST /summarise (with arxiv_id), or GET /papers/{arxiv_id}/related.

Parameters

Try it out

Name	Description
arxiv_id * required string (path)	arxiv_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema	*No links*
401	Missing or invalid API key	*No links*
404	Paper has not been interacted with yet — not in the community index	*No links*

```
{
  "arxiv_id": "1706.03762",
  "authors": "Ashish Vaswani, Noam Shazeer",
  "categories": "cs.CL cs.LG",
  "interaction_count": 14,
  "last_interacted_at": "2025-03-10T11:42:00Z",
  "title": "Attention Is All You Need",
  "url": "https://arxiv.org/abs/1706.03762",
  "year": 2017
}
```

422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


Notes

Public notes on arXiv papers — readable by all consumers and agents. Full CRUD. Write endpoints rate limited to 10 requests/minute.

POST `/papers/{arxiv_id}/notes` Add a public note to a paper

Add a public note to an arXiv paper. Notes are visible to all API consumers and agents.

Parameters **Try it out**

Name	Description
arxiv_id * required `string` *(path)*	arxiv_id

Request body required **application/json**

Example Value Schema

```
{
  "content": "Key contribution: introduces multi-head attention as a drop-in r
eplacement for RNNs."
}
```

Responses

Code	Description	Links
201	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema `{ "arxiv_id": "1706.03762", "content": "Key contribution: introduces multi-head a ttention as a drop-in replacement for RNNs.", "created_at": "2025-03-10T10:00:00Z", "id": 1, "updated_at": "2025-03-10T10:00:00Z" }`	*No links*
401	Missing or invalid API key	*No links*
404	Paper not found in corpus	*No links*
422	Validation error — content is required	*No links*
429	Rate limit exceeded — max 10 notes per minute	*No links*

GET /papers/{arxiv_id}/notes Get all notes for a paper

Retrieve all public notes attached to a specific arXiv paper.

Parameters

Try it out

Name	Description
arxiv_id * required string (path)	arxiv_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. **Example Value** Schema	*No links*

```
[
  {
    "arxiv_id": "1706.03762",
    "content": "Key contribution: introduces multi-head
attention as a drop-in replacement for RNNs.",
    "created_at": "2025-03-10T10:00:00Z",
    "id": 1,
    "updated_at": "2025-03-10T10:00:00Z"
  }
]
```

401	Missing or invalid API key	*No links*
404	Paper not found in corpus	*No links*
422	Validation Error Media type application/json **Example Value** Schema	*No links*

```
{
```

```
    "detail": [
      {
        "loc": [
          "string",
          0
        ],
        "msg": "string",
        "type": "string",
        "input": "string",
        "ctx": {}
      }
    ]
  }
```


PATCH /notes/{note_id} Update a note

Update the content of an existing note by its ID.

Parameters

Try it out

Name	Description
note_id * required integer (path)	note_id

Request body required

application/json

Example Value Schema


```
{
  "content": "Updated: also introduces positional encoding to preserve sequence order."
}
```


Responses

Code	Description	Links

200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "arxiv_id": "1706.03762",
  "content": "Key contribution: introduces multi-head a
ttention as a drop-in replacement for RNNs.",
  "created_at": "2025-03-10T10:00:00Z",
  "id": 1,
  "updated_at": "2025-03-10T10:00:00Z"
}
```


401	Missing or invalid API key	*No links*
404	Note not found	*No links*
422	Validation error — content is required	*No links*
429	Rate limit exceeded — max 10 updates per minute	*No links*

DELETE /notes/{note_id} Delete a note

Permanently delete a note by its ID.

Parameters

Try it out

Name	Description
note_id * required `integer` *(path)*	note_id

Responses

Code	Description	Links
204	Successful Response	*No links*
401	Missing or invalid API key	*No links*
404	Note not found	*No links*
422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Utility System health and status checks.

GET **/health** Health check

Check the API is running and responsive.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema `"string"`	*No links*

Schemas